ELEC 221 Lecture 10
DT filters; Introducing the Fourier transform

Tuesday 13 October 2026

Announcements

- Quiz 5 today
- Midterm in 3 weeks. Practice tests will be posted on PrairieLearn by end of week.

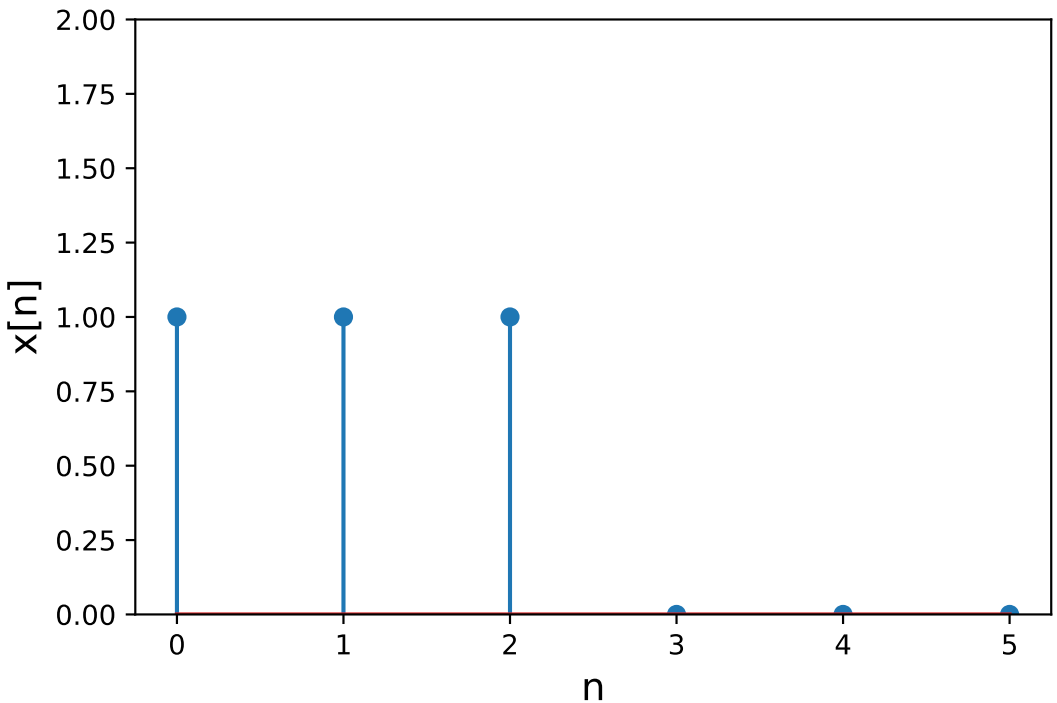
Last time

DT Fourier synthesis equation

DT Fourier analysis equation

Last time

We computed Fourier coefficients of a DT square wave



Today

Learning outcomes:

- Apply properties of DT Fourier series to determine coefficients of transformed signals
- Distinguish between finite impulse response and infinite impulse response filters in DT
- Distinguish between the CT Fourier series and Fourier transform
- Compute the Fourier spectrum of a CT signal

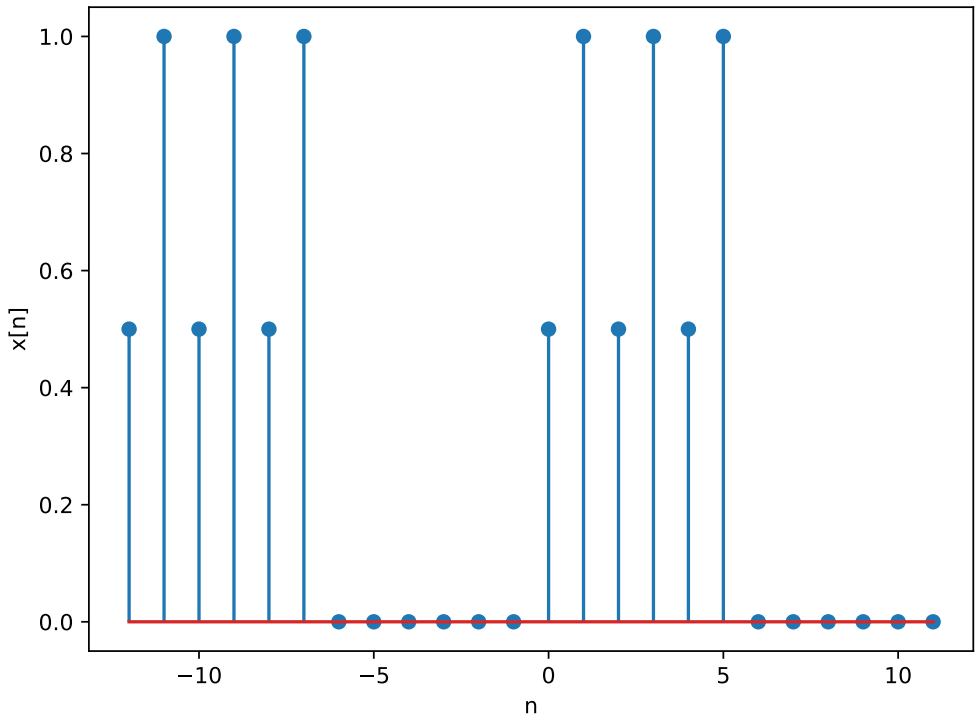
Properties of DT Fourier coefficients

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$x[n]$, $y[n]$: Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	a_k, b_k: Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m}a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^{n} x[k]$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$a_k = a^*_{-k}$; $\mathcal{R}e\{a_k\} = \mathcal{R}e\{a_{-k}\}$; $\mathcal{I}m\{a_k\} = -\mathcal{I}m\{a_{-k}\}$; $\lvert a_k\rvert = \lvert a_{-k}\rvert$; $\measuredangle a_k = -\measuredangle a_{-k}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}v\{x[n]\}$ [x[n] real]; $x_o[n] = \mathcal{O}d\{x[n]\}$ [x[n] real]	$\mathcal{R}e\{a_k\}$; $j\mathcal{I}m\{a_k\}$

Exercise: DT Fourier coefficients

How would we compute the Fourier coefficients for this signal?



Exercise: the DT square wave

DT filters

Recall that in DT, the frequency increases up until $\omega = \pi$, then decreases as it approaches 2π.

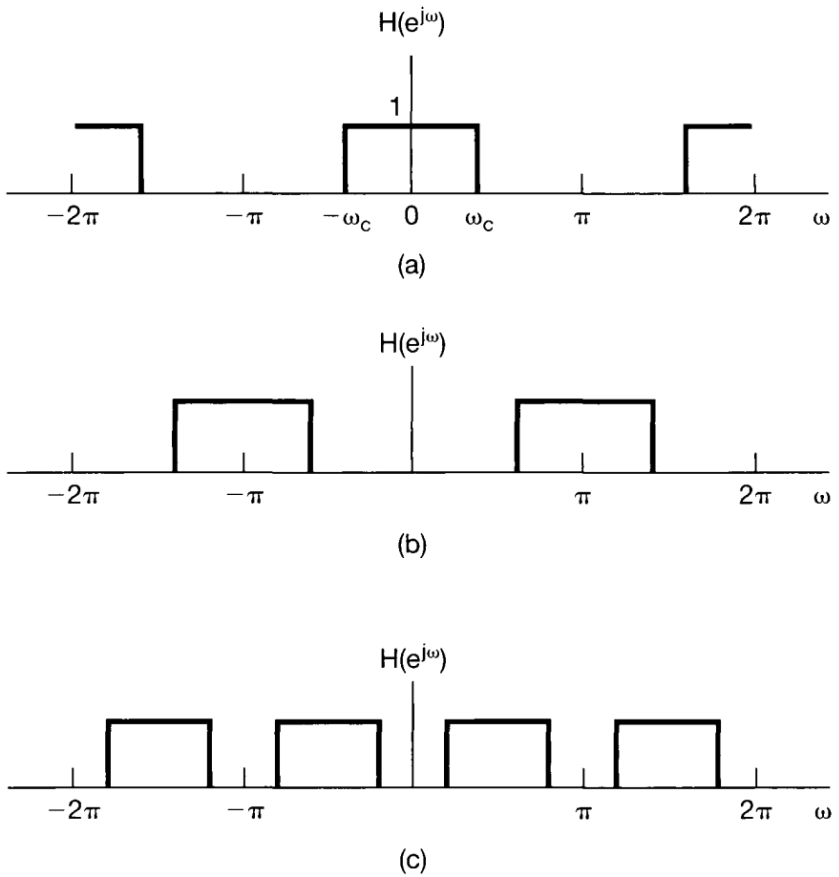
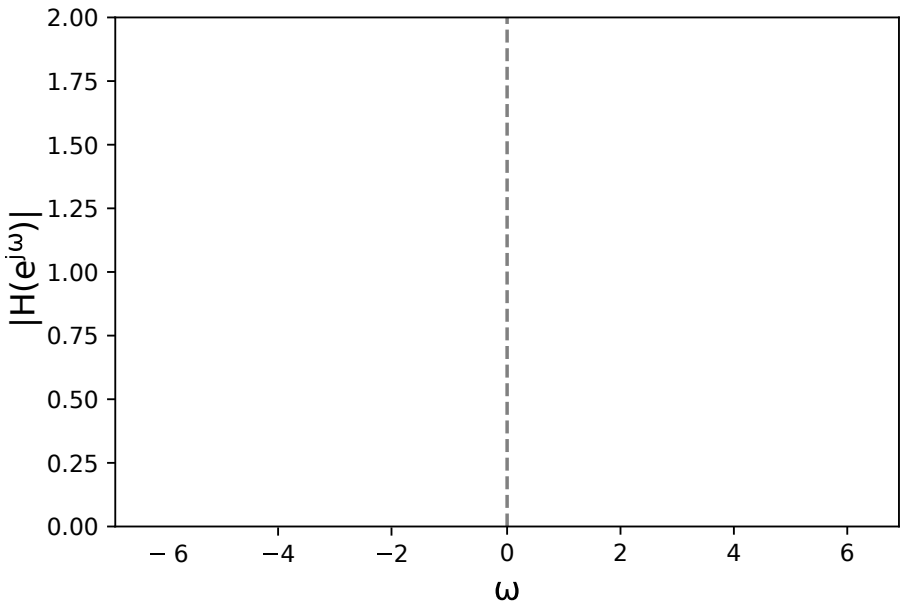


Image credit: Oppenheim chapter 3.9.

DT filters

Example: consider the DT filter with frequency response

Sketch below:



DT filters

There are two major categories of DT filters:

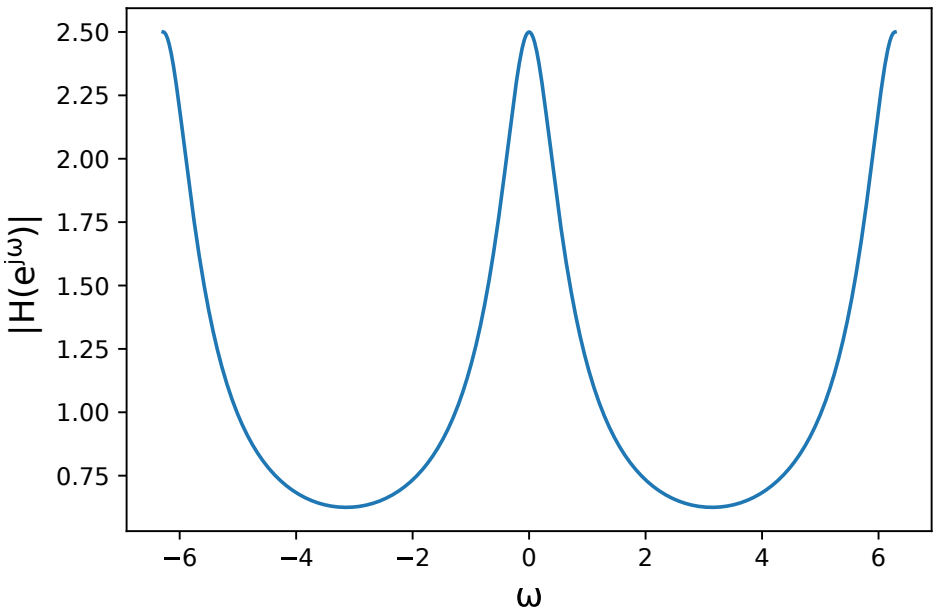
1. Infinite impulse response (IIR)
2. Finite impulse response (FIR)

DT filters (IIR)

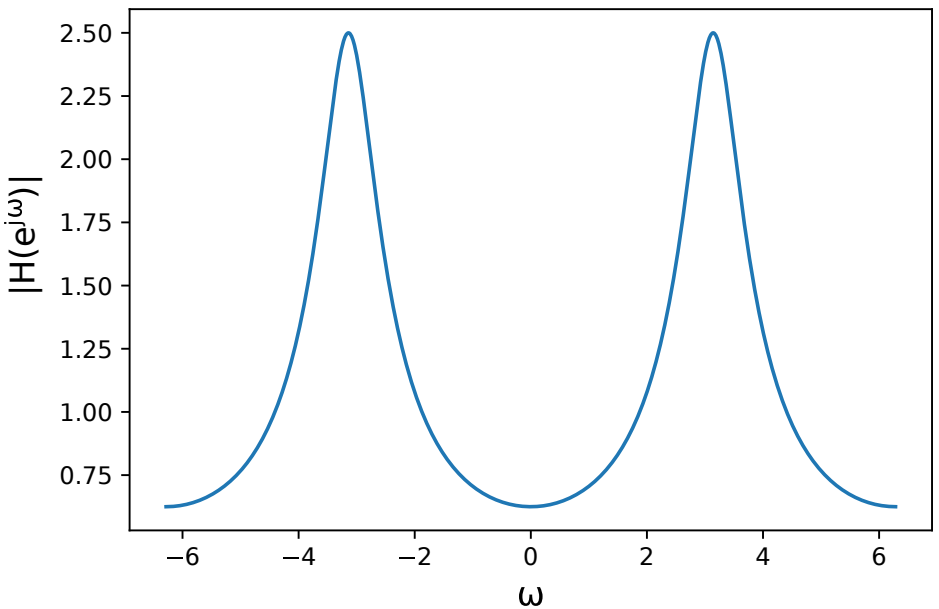
Example: we can generate a low- or high-pass filter using a system described by a (recursive) first-order difference equation.

DT filters (IIR)

Case: $a > 0$



Case: $a < 0$



DT filters (IIR)

To determine the impulse response, assume the system starts at rest ($y[-1] = 0$) and solve recursively:

This system has infinite impulse repsonse:

DT filters (FIR)

An example of a DT filter with finite impulse response is the moving average with window of size M:

Impulse response: set $x[n] = \delta[n]$

This is clearly finite (as a result such filters are stable).

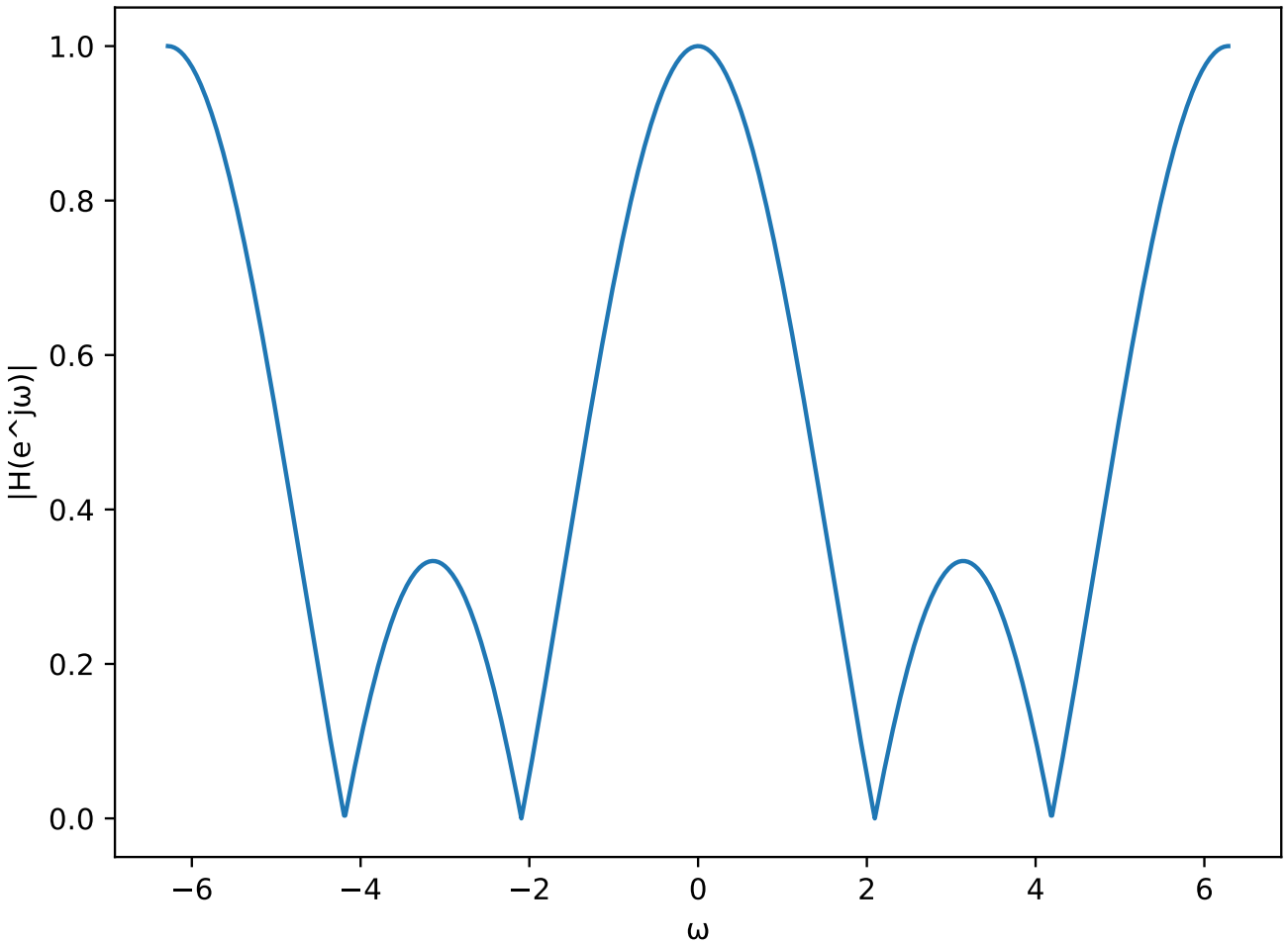
DT filters (FIR)

Let's compute the frequency response:

Set $M = 1$ (3-point moving average)

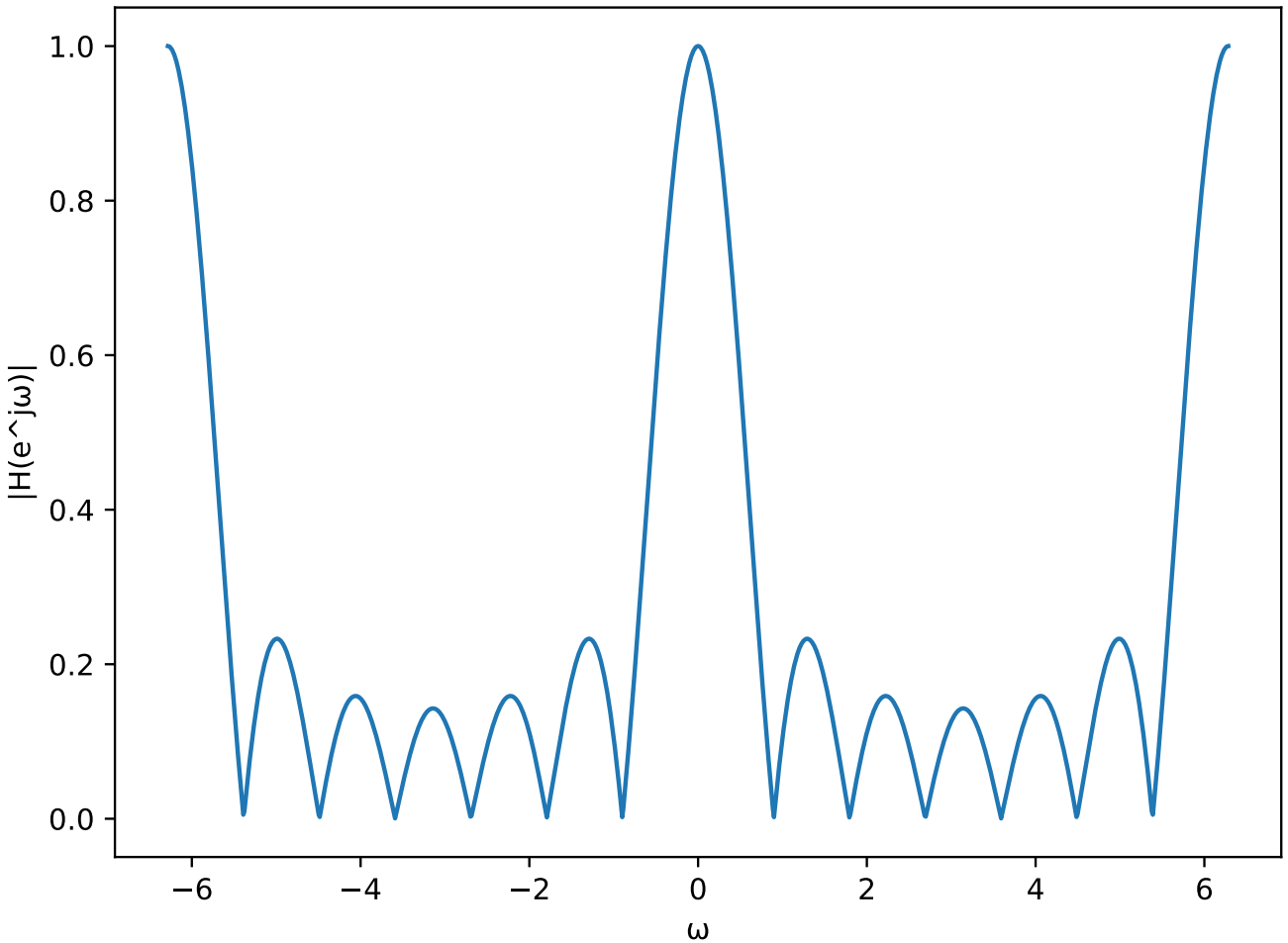
DT filters (FIR)

Example when $M = 1$



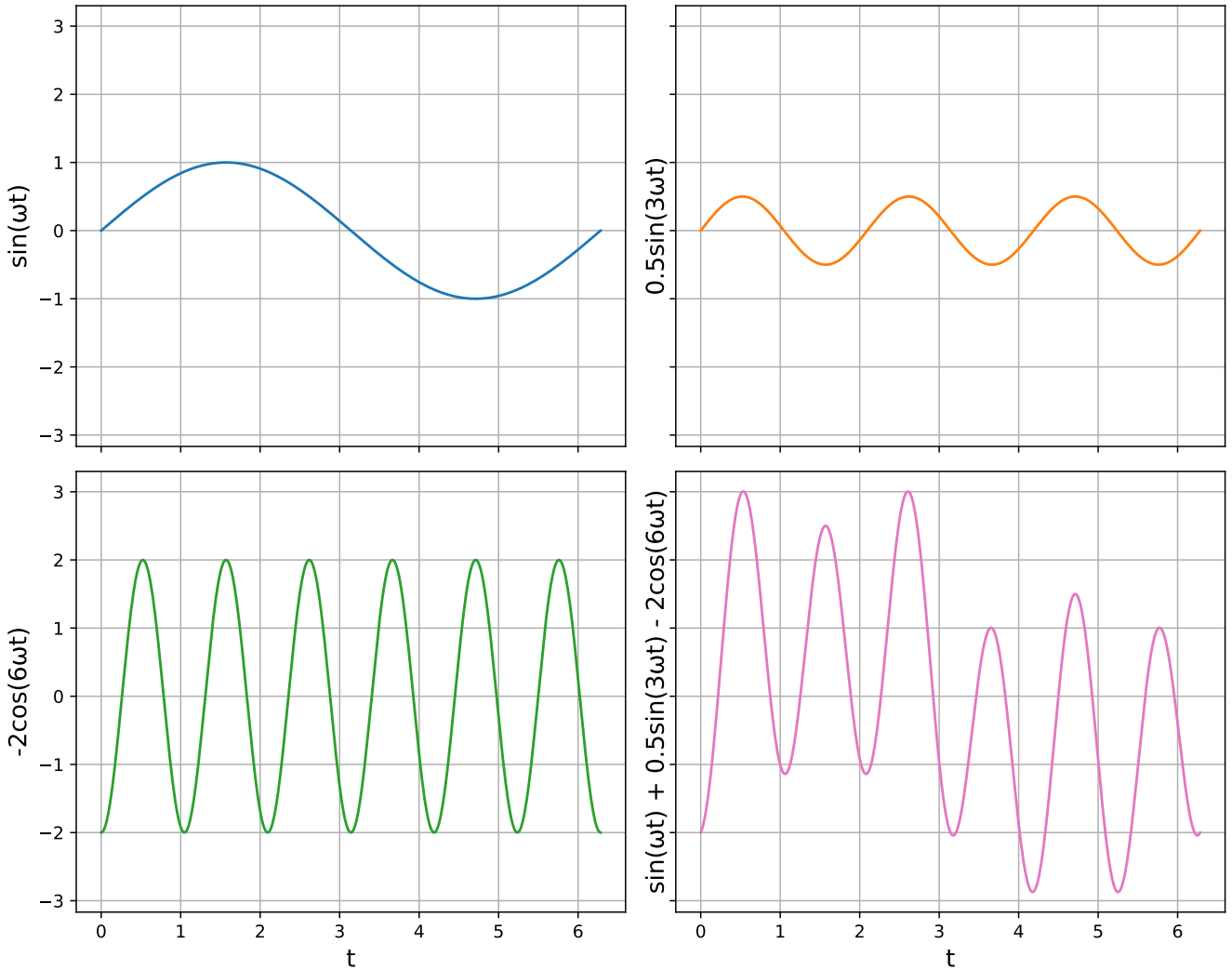
DT filters (FIR)

Example when $M = 3$



Fourier series

We've seen the Fourier series representation of **periodic** signals:



Fourier series

CT synthesis equation:

CT analysis equation:

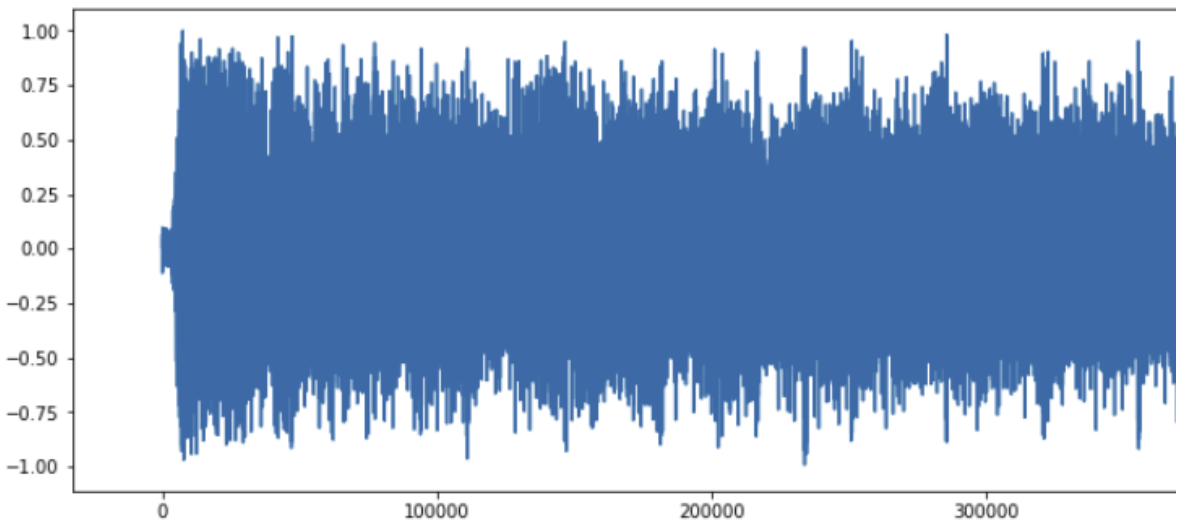
A periodic signal is composed of complex exponential signals with *integer multiples of the fundamental frequency* only (harmonics).

Fourier series

In tutorial assignment 2, you worked with signals that *weren't periodic*:

```
plt.plot(audio)
```

```
[<matplotlib.lines.Line2D at 0x7f940880e370>]
```



But, we were still doing *something* with Fourier analysis:

```
fourier_spectrum = np.fft.rfft(audio)
```

The Fourier transform

The **Fourier transform** generalizes the Fourier series to **aperiodic signals**. It involves all possible frequencies.

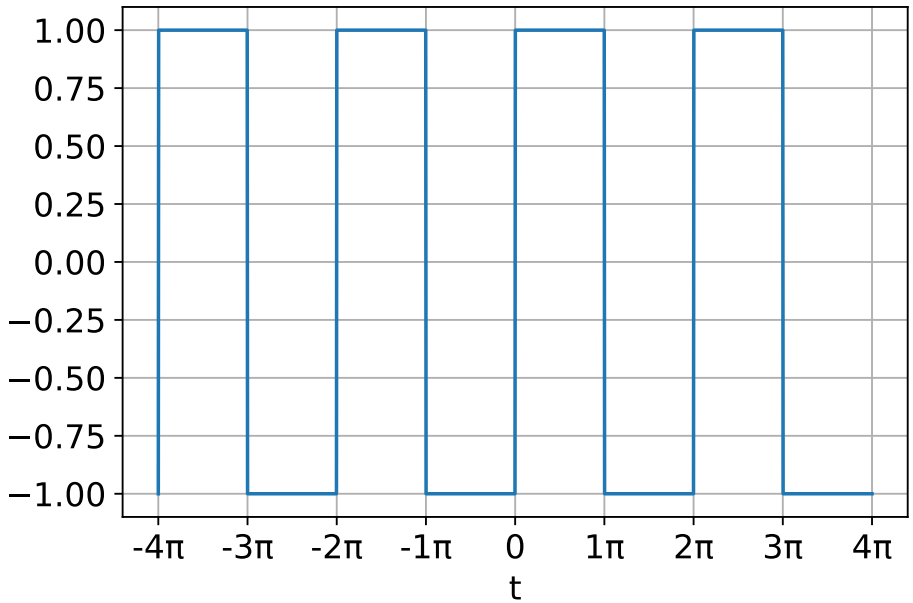
Fourier series:

Fourier transform:

How do we get here?

Towards the Fourier transform

Previously, we looked at a 2π-periodic square wave:



We derived its Fourier series representation

Towards the Fourier transform

Let's generalize this. Consider the following square wave:

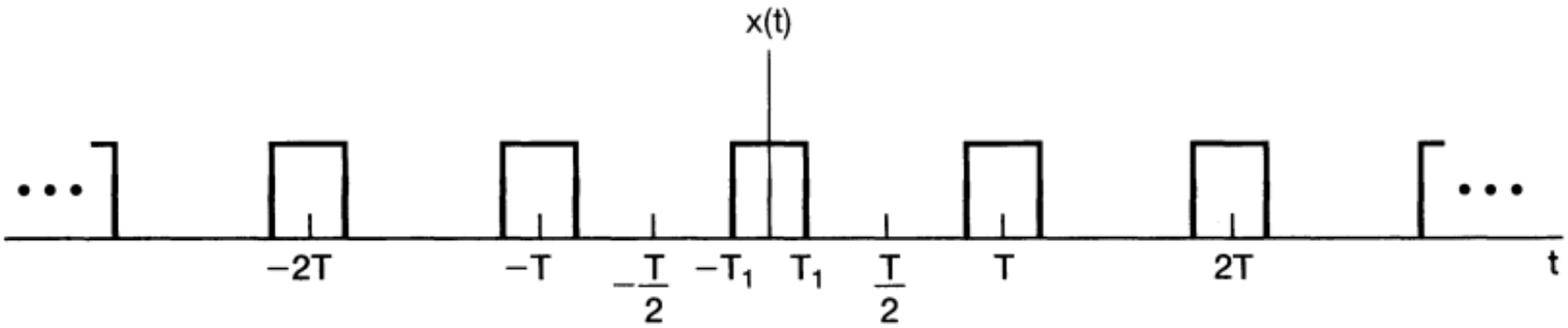


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Let's compute its Fourier coefficients.

$$c_k = \frac{1}{T} \int_T x(t) e^{-jk\omega t}$$

Start with c_0:

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Now the c_k:

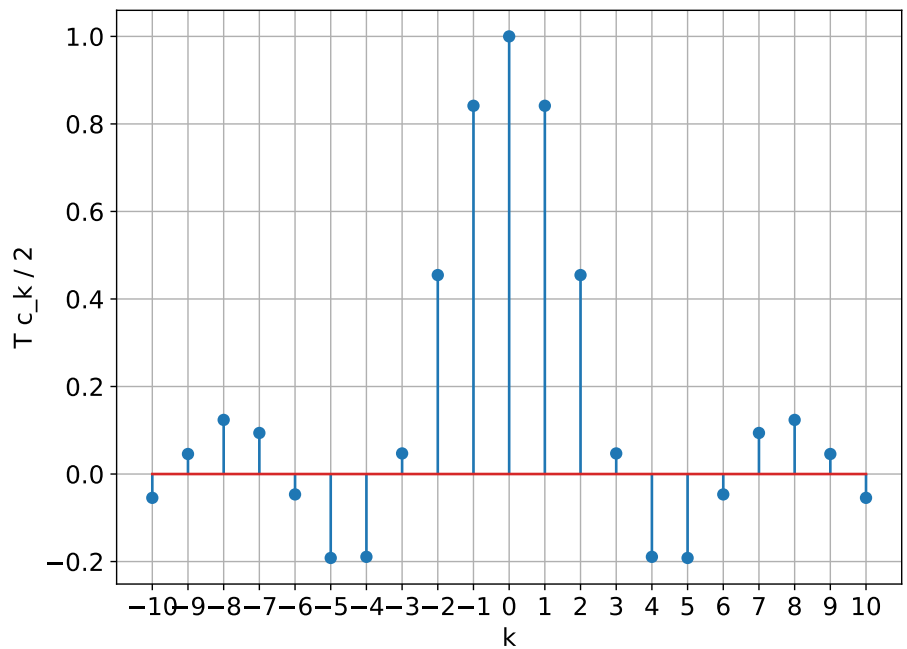
Towards the Fourier transform

What does this function look like?

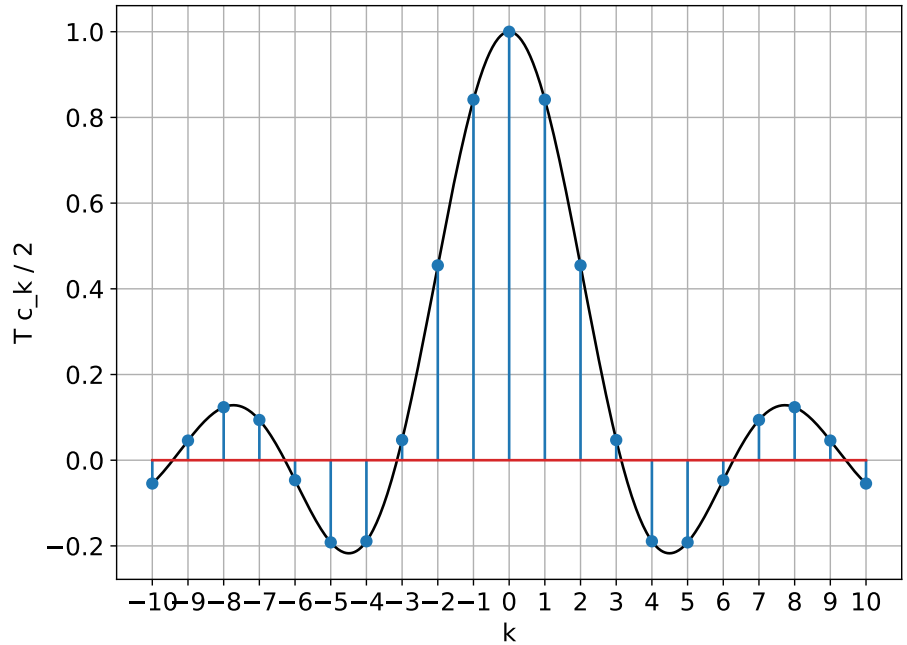
Rearrange and express as a function of k:

Let's plot this: set $T_1 = 1$, and $T = 2\pi$, so $\omega = \frac{2\pi}{T} = 1$.

Towards the Fourier transform



Towards the Fourier transform



These are **samples** of the function

at *integer values of* k.

Towards the Fourier transform

Let's consider this instead as a function of frequency, $\tilde{\omega} = k\omega$:

The Fourier coefficients are samples of this function at *integer multiples* of fundamental frequency, $\tilde{\omega} = k\omega$, where $\omega = 2\pi/T$.

Towards the Fourier transform

Suppose T grows, but T_1 stays the same:

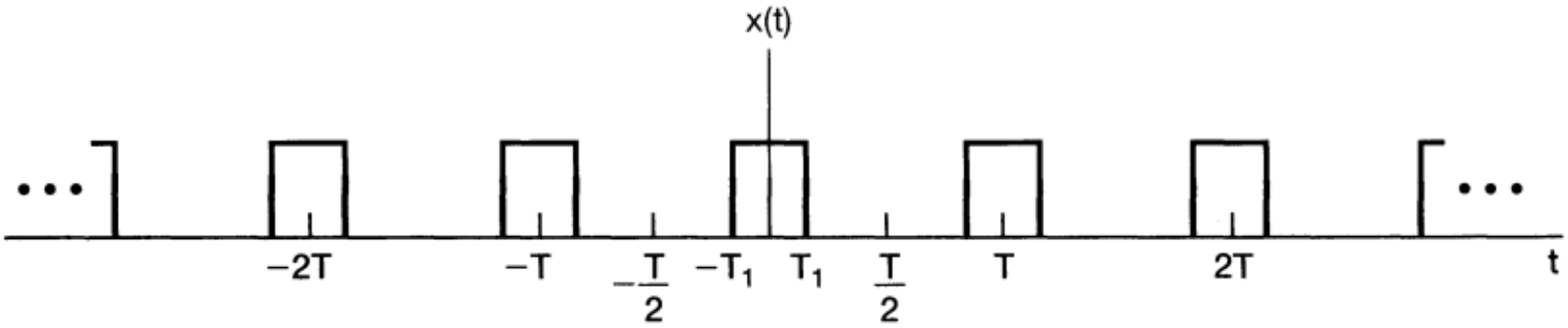


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

Initially the spacing of samples is integer multiples of $\omega = 2\pi/T$.

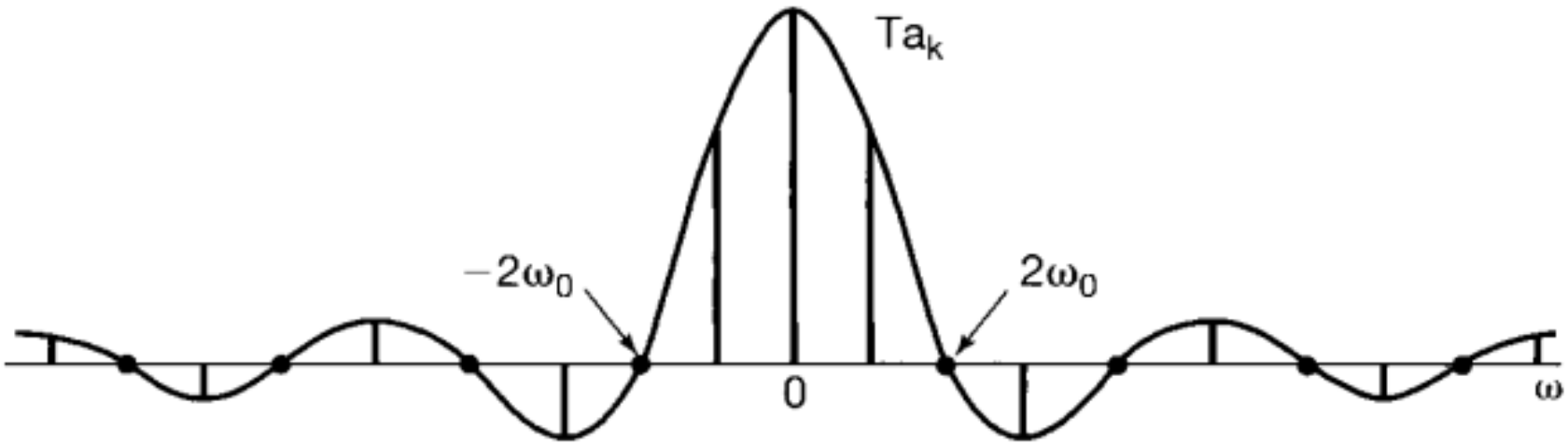
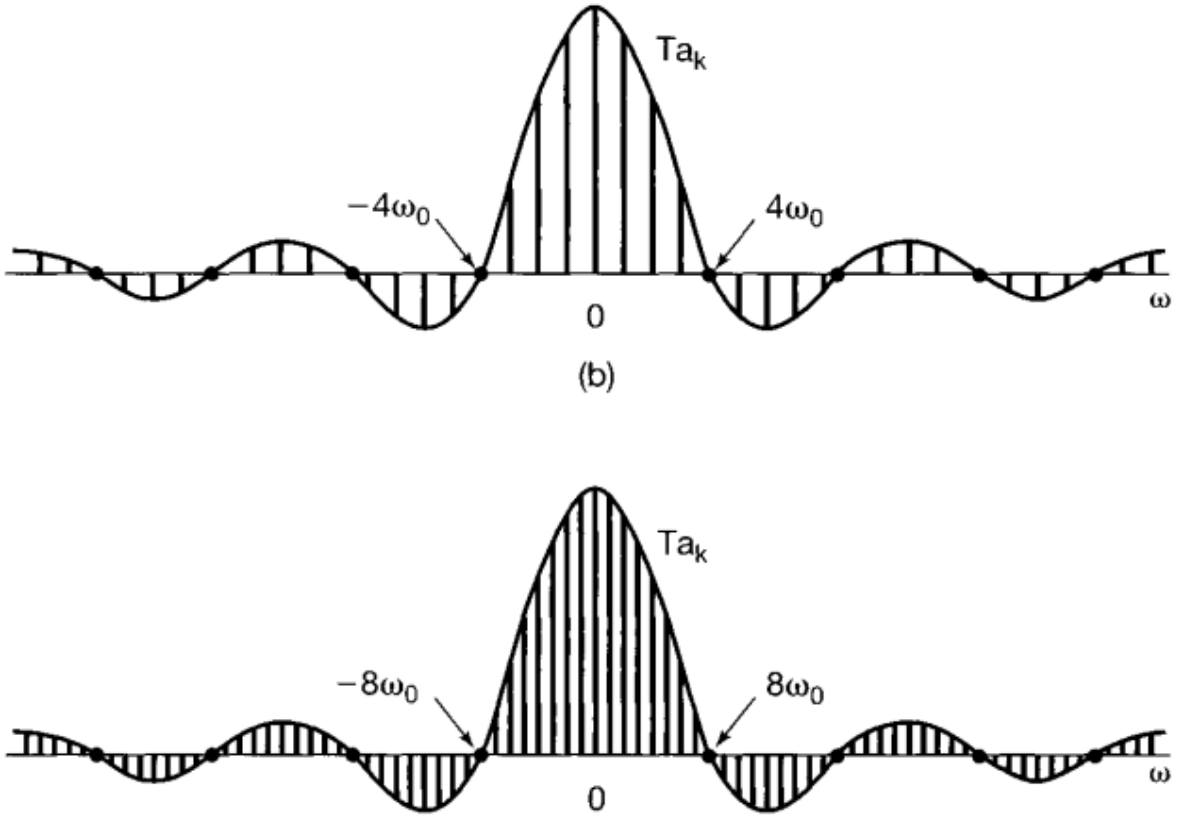


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

As T grows, $\omega = 2\pi/T$ becomes smaller and smaller, so the integer multiples of it get closer and closer together.



Towards the Fourier transform

Eventually, ω becomes so small that instead of

we may as well just consider the sum over integer multiples as a continuous integral over all possible ω:

...but what does this have to do with non-periodic signals?

Towards the Fourier transform

Given any aperiodic signal $x(t)$, we can always "pretend" it's periodic by constructing a **periodic extension**, $\tilde{x}(t)$ with period T.

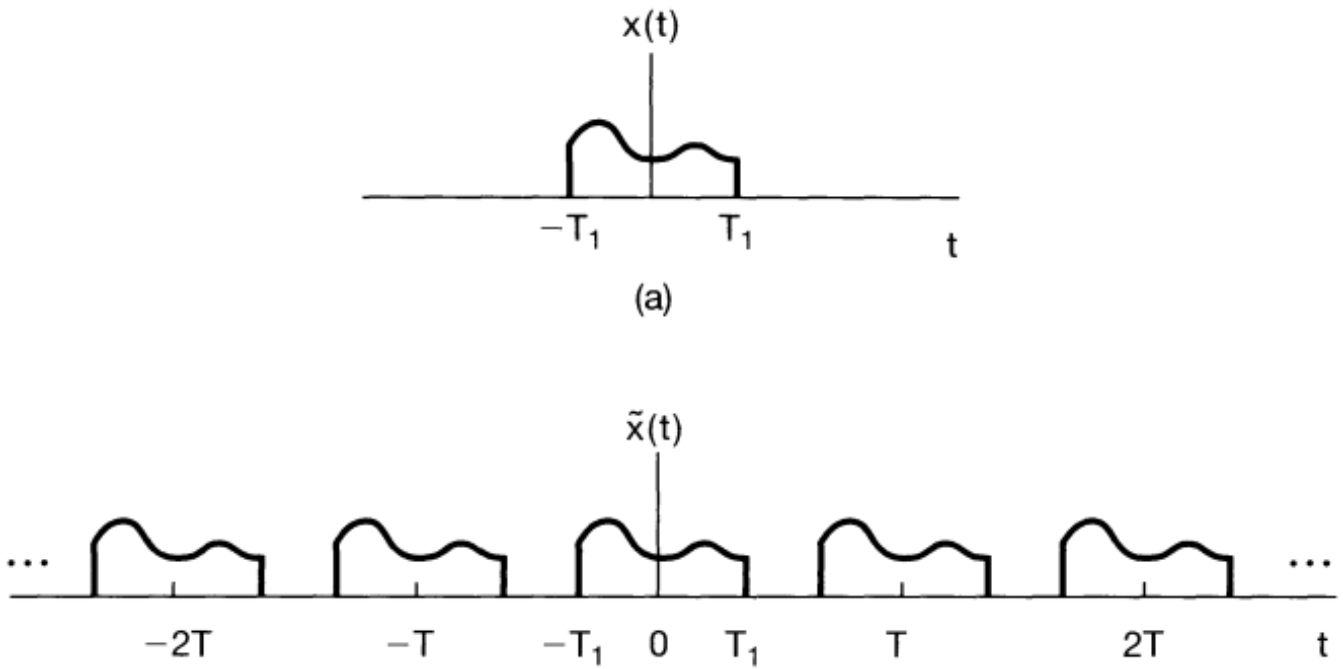


Image credit: Oppenheim chapter 4.1

Motivation: Fourier transform

We can express $\tilde{x}(t)$ as a Fourier series (where $\omega = 2\pi/T$):

Motivation: Fourier transform

What happens to the coefficients?

Let's define

so that

Motivation: Fourier transform

We can put this back in our Fourier series:

Motivation: Fourier transform

Consider what happens when $T \to \infty$...

1. $\tilde{x}(t)$ will look just like $x(t)$ for large enough T

2. ω will get smaller and smaller

The Fourier transform

Inverse Fourier transform (synthesis equation):

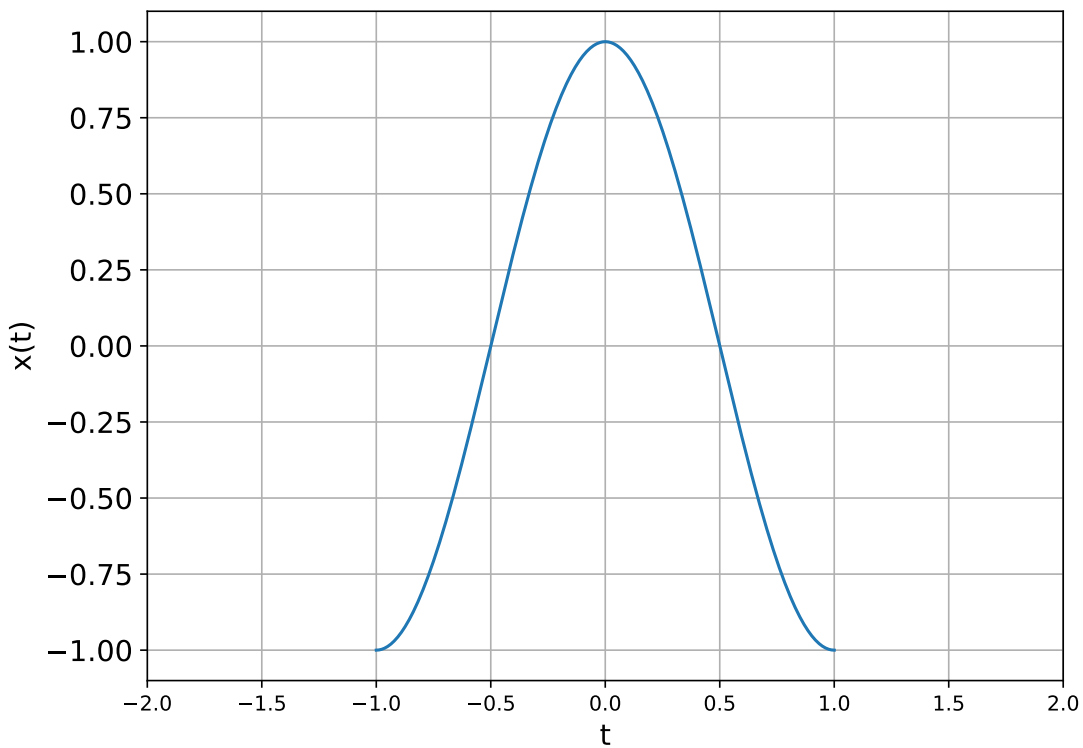
Fourier transform (analysis equation, or Fourier *spectrum*):

Note: Sometimes the $1/2\pi$ prefactor appears on the spectrum, or sometimes both versions have $1/\sqrt{2\pi}$.

Example

Compute the Fourier spectrum of:

$$x(t) = \begin{cases} \cos(\pi t), & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$



Example

$$x(t) = \begin{cases} \cos(\pi t), & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$

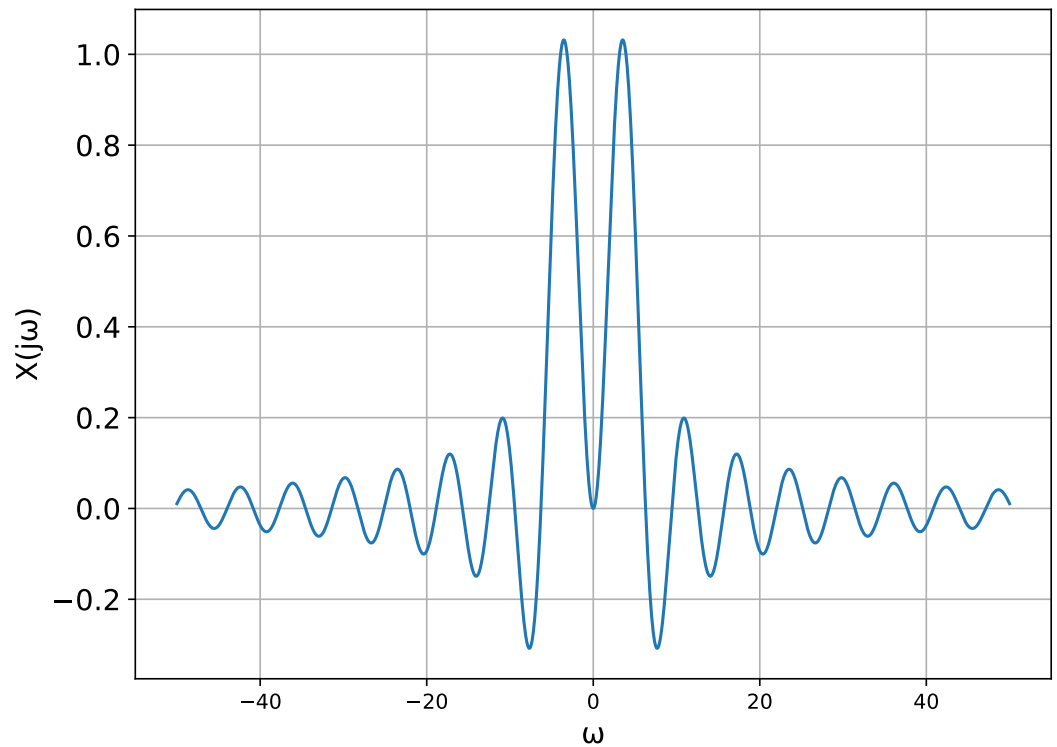
Start from the definition:

Example

$$X(j\omega) \qquad \frac{1}{2}\int_{-1}^{1} e^{j(\pi-\omega)t}dt + \frac{1}{2}\int_{-1}^{1} e^{-j(\pi+\omega)t}dt$$

Example

$$X(j\omega) = \frac{\sin(\omega)}{\pi - \omega} - \frac{\sin(\omega)}{\pi + \omega}$$



Fourier transform and impulse response

You've actually already seen the Fourier transform before...

Write a signal as a combination of shifted, weighted impulses:

Put this in an LTI system with impulse response $h(t)$:

Fourier transform and impulse response

$$x(t) \to y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

When the signal in question is a complex exponential,

Fourier transform and impulse response

The system function $H(j\omega)$, or frequency response

is the **Fourier transform of the impulse response**!

We can use the inverse Fourier transform to obtain the impulse response from the frequency response:

For next time

Content:

- Fourier transform for periodic signals
- Properties of the CT Fourier *transform*
- Time/frequency duality

Action items:

1. Work on practice problems

Recommended reading:

- From today's class: Oppenheim 3.7, 3.11, 4.0-4.2
- Suggested problems: 4.1, 4.2a, 4.21abei, 4.22abde
- For next class: Oppenheim 4.2-4.4